

COSC 594

DM for Robotic Planning (II)

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Outline

- RECAP
- DIFFUSION MODEL MOTION PLANNING
- DIFFUSION MODEL COMPOSITION PERSPECTIVE

Outline

■ RECAP

Motivation: why robot learning?

■ END-TO-END ROBOTICS



Gemini Robotics 1.5

- ★ learn from data/experience
- ★ multiple tasks
- ★ dynamic environment
- ★ cost-effective and quick design

Planning as inference

Random trajectory: $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$

Posterior trajectory distribution: $p(\tau | h) \propto p(h | \tau)p(\tau)$

$p(h | \tau)$ – likelihood of completing tasks

$p(\tau)$ – prior

■ MAXIMUM A POSTERIORI

maximize _{τ} $p(\tau | h)$



maximize _{τ} $\log p(\tau | h)$

Sampling from energy-based model

Energy-based model: $p_{\theta}(h | \tau) \propto \prod_{k=1}^K e^{E_{\theta^k}(\tau; h^k)}$

Prior distribution: $p(\tau) \propto p(s_0) p(s_T) e^{\|\tau - \mu_{\tau}\|^2}$

■ POSTERIOR TRAJECTORY DISTRIBUTION

$$p(\tau | h) \propto p_{\theta}(h | \tau) \underbrace{p(\tau)}_{\text{Gaussian prior}} \propto e^{E_{\theta}(\tau; h)}$$

★ Langevin dynamics

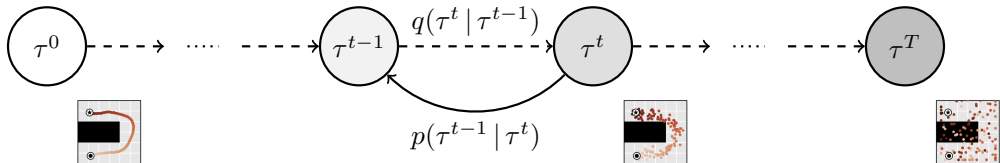
$$\tau_t = \tau_{t-1} + \eta \nabla_{\tau} E_{\theta}(\tau_{t-1}; h) + \sqrt{2\eta} \epsilon_t \quad \text{where } \epsilon_t \sim \mathcal{N}(0, I)$$

As $t \rightarrow \infty$, dynamics converge to samples $\tau_t \sim p(\cdot | h)$

Outline

- MOTIVATION
- DIFFUSION MODEL MOTION PLANNING

Diffusion model as trajectory generative model



data point τ^0 – trajectory

■ ANALOGUE W/ VAE

★ noiser (encoder)

$$q(\tau^{1:T} | \tau^0) = q(\tau^1 | \tau^0) q(\tau^2 | \tau^1) \dots q(\tau^T | \tau^{T-1})$$

★ denoiser (decoder)

$$p(\tau^0 | \tau^{1:T}) = p(\tau^T) p(\tau^{T-1} | \tau^T) \dots p(\tau^0 | \tau^1)$$

■ GROUND-TRUTH DENOISER

$$q(\tau^{t-1} | \tau^t, \tau^0) = \mathcal{N}(\tau^{t-1}; \mu(\tau^t, \tau^0), \sigma^2(t)I)$$

$$q(\tau^t | \tau^{t-1}) = \mathcal{N}(\tau^t; \sqrt{\alpha_t} \tau^{t-1}, (1 - \alpha_t)I)$$

$$\mu(\tau^t, \tau^0) = \frac{1}{\sqrt{\alpha_t}} \tau^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \epsilon_0 - \text{noise-based mean}$$

$$\sigma^2(t) - \text{variance}$$

■ APPROXIMATE DENOISER

$$p_\theta(\tau^{t-1} | \tau^t) = \mathcal{N}(\tau^{t-1}; \mu_\theta(\tau^t, t), \sigma^2(t)I), \quad \mu_\theta(\tau^t, t) = \frac{1}{\sqrt{\alpha_t}} \tau^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \epsilon_\theta(\tau^t, t)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(\tau^{t-1} | \tau^t, \tau^0) \| p_\theta(\tau^{t-1} | \tau^t)) \Leftrightarrow \underset{\theta}{\text{minimize}} \quad \|\epsilon_0 - \epsilon_\theta(\tau^t, t)\|^2$$

Cloned trajectory $p_{\theta^\star}(\tau^{t-1} | \tau^t)$

Conditional diffusion model

Conditional diffusion model: $p_{\theta}(\tau^{t-1} | \tau^t, h)$ w/ task info h

■ BACKWARD GENERATIVE PROCESS

$$p_{\theta}(\tau^{0:T} | h) = p(\tau^T | h) \prod_{t=1}^T p_{\theta}(\tau^{t-1} | \tau^t, h)$$

$p(\tau^T | h)$ – standard Gaussian

★ task-conditioned posterior

$$p_{\theta}(\tau^{t-1} | \tau^t, h) \propto \underbrace{p_{\theta}(h | \tau^t, \tau^{t-1})}_{\text{task likelihood } p(h | \tau^{t-1})} \times \underbrace{p_{\theta}(\tau^{t-1} | \tau^t)}_{\text{unconditional model/prior}}$$

Sampling from conditional diffusion model

Conditional diffusion model: $p_{\theta}(\tau^{t-1} | \tau^t, h)$ w/ task info h

■ UNCONDITIONAL MODEL/PRIOR

$$\log p_{\theta}(\tau^{t-1} | \tau^t) = \log \mathcal{N}(\tau^{t-1}; \mu_{\theta}(\tau^t, t), \sigma^2(t)I) \approx -\frac{1}{2} \|\tau^{t-1} - \mu_{\theta}(\tau^t, t)\|^2$$

■ TASK LIKELIHOOD

$$\log p_{\theta}(h | \tau^{t-1}) \approx \log p(h | \tau^{t-1} = \mu_{\theta}(\tau^t, t)) + \underbrace{(\tau^{t-1} - \mu_{\theta}(\tau^t, t)) \nabla_{\tau} \log p(h | \tau^{t-1})|_{\tau^{t-1} = \mu_{\theta}(\tau^t, t)}}_{\text{guidance } g}$$

■ TASK-CONDITIONED POSTERIOR

$$\begin{aligned}\log p_{\theta}(\tau^{t-1} \mid \tau^t, h) &\approx \log p_{\theta}(\tau^{t-1} \mid \tau^t) + \log p_{\theta}(h \mid \tau^t, \tau^{t-1}) \\&= -\frac{1}{2} \|\tau^{t-1} - \mu_{\theta}(\tau^t, t)\|^2 + (\tau^{t-1} - \mu_{\theta}(\tau^t, t)) g \\&\approx -\frac{1}{2} \|\tau^{t-1} - \mu_{\theta}(\tau^t, t) - g\|^2 \\&\quad \Downarrow \\p_{\theta}(\tau^{t-1} \mid \tau^t, h) &= \mathcal{N}(\tau^{t-1}; \mu_{\theta}(\tau^t, t) + g, \bar{\sigma}^2(t)I)\end{aligned}$$

How to compute guidance $g = \nabla_{\tau} \log p(h \mid \tau^{t-1})|_{\tau^{t-1}=\mu_{\theta}(\tau^t, t)}$?

Planning reward functions

Guidance: $g = \nabla_{\tau} \log p(h \mid \tau^{t-1})|_{\tau^{t-1}=\mu_{\theta}(\tau^t, t)}$

■ INITIAL/FINAL STATES

$$p(h \mid \tau) = e^{-\|s_0 - \tau[0]\|^2} \implies \nabla_{\tau} \log p(h \mid \tau^{t-1}) = -\|s_0 - \tau[0]\|^2$$

$$p(h \mid \tau) = e^{-\|s_T - \tau[T]\|^2} \implies \nabla_{\tau} \log p(h \mid \tau^{t-1}) = -\|s_T - \tau[T]\|^2$$

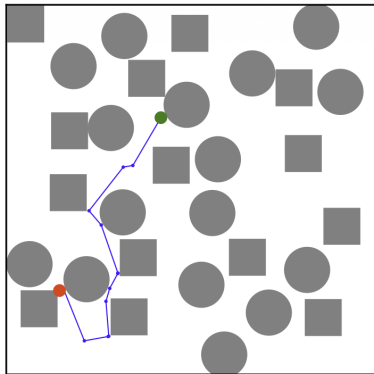
■ COLLISION COST

$$p(h \mid \tau) = e^{-c(\tau)} \implies \nabla_{\tau} \log p(h \mid \tau^{t-1}) = -c(\tau)$$

Differentiable over trajectory $\implies$ **tractable to compute** $g = \nabla_{\tau} \log p(h \mid \tau)$

Experiment I: 2D maze

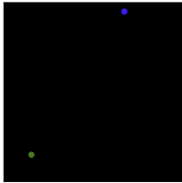
POINT-MASS2D DENSE



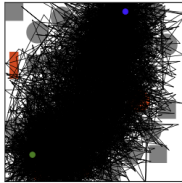
Random start/goal positions

Experiment I: 2D maze

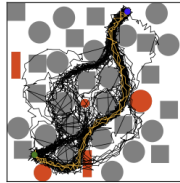
■ DIFFUSION TRAJECTORIES



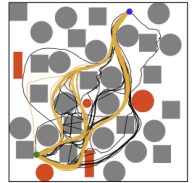
(a) Diffusion Step 0



(b) Diffusion Step 10



(c) Diffusion Step 20

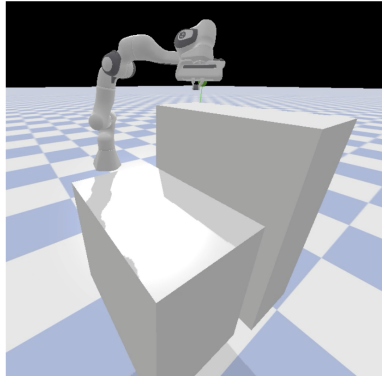


(d) Diffusion Step 25

Feasible trajectories under **new obstacles (red)**

Experiment II: Panda robot

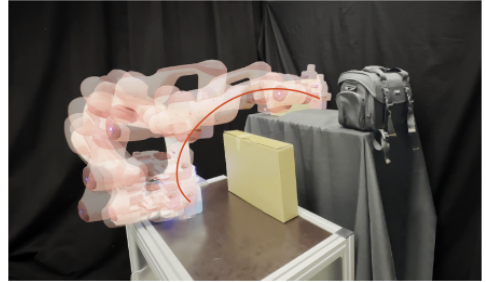
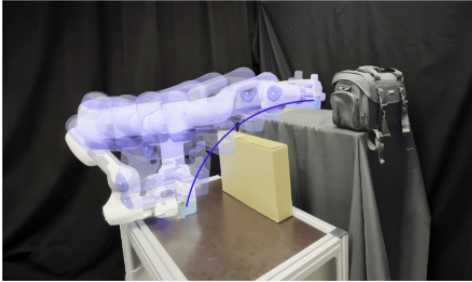
■ PANDA SHELF ROBOT



Expert data trajectories

Experiment II: 2D maze

■ DIFFUSION TRAJECTORIES



Feasible trajectories (w/ two modes) under **a new obstacle (box)**

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- DIFFUSION MODEL COMPOSITION PERSPECTIVE

Diffusion model composition for planning

K diffusion models: $\mu_{\theta^k}(\tau^t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(\tau^0)$ for $k = 1, \dots, K$

Reward model: $r(\tau^0)$ for a sample τ^0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{\tau^0 \sim p(\cdot)}[r(\tau^0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot))$$

$\beta_1, \dots, \beta_K$ – tradeoff parameters

★ optimal model $p^\star$

$$p^\star(\tau^0) \propto p^{\text{con}}(\tau^0) e^{r(\tau^0)/\mathbf{1}^\top \beta} \quad \text{and} \quad p^{\text{con}}(\tau^0) = \prod_{k=1}^K (p^k(\tau^0))^{\beta_k / \mathbf{1}^\top \beta} \times \frac{1}{Z}$$

Exponentially weighted conjunction distribution

Diffusion model composition as inference

K diffusion models: $\mu_{\theta^k}(\tau^t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(\tau^0)$ for $k = 1, \dots, K$

Reward model: $r(\tau^0)$ for a sample τ^0

■ BAYESIAN PERSPECTIVE

$$p^*(\tau^0) \propto \underbrace{p^{\text{con}}(\tau^0) e^{r(\tau^0)/\mathbf{1}^\top \beta}}_{\text{evidence}} \quad \text{and} \quad \underbrace{p^{\text{con}}(\tau^0) = \prod_{k=1}^K \left(p^k(\tau^0) \right)^{\beta_k / \mathbf{1}^\top \beta}}_{\text{prior}} \times \frac{1}{Z}$$

Optimal model p^* is a posterior distribution

Summary

■ RECAP

■ DIFFUSION MODEL MOTION PLANNING

Trajectory generative model, conditional diffusion model

■ DIFFUSION MODEL COMPOSITION PERSPECTIVE

Model composition-based planning